

Revised Load Interconnection System Impact Study Report

For Queue Project Q22-1979

McQueeney 138 kV
October 29, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-1979. The Interconnection Customer proposes a 255.5 MW large load (data center) facility interconnecting at the McQueeney 138 kV substation in the Virginia area. The request was received on January 10, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 521 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$20,912,000.

3. Network Impacts

3.1 Load Deliverability Analysis

1. Facility: Dona Vista - McQueeney 138.0 kV Line
Contingency: Los Angeles 1617 - Hartselle 138.0 kV
Loading: 165.3% (Rating: 188 MVA)
DFAX: 99.9%
Pre-project loading: 29.3%
2. Facility: Villa Mar 1657 - McQueeney 138.0 kV
Contingency: Works 53 - Mickens 345.0 kV
Loading: 161.8% (Rating: 214 MVA)
DFAX: 91.3%
Pre-project loading: 52.7%
3. Facility: Los Angeles 1617 - Hartselle 138.0 kV
Contingency: Works 53 - Mickens 345.0 kV
Loading: 157.6% (Rating: 242 MVA)
DFAX: 91.2%
Pre-project loading: 61.4%
4. Facility: Villa Mar 1657 - Los Angeles 1617 Substation 138.0 kV L
Contingency: Works 53 - Mickens 345.0 kV L
Loading: 144.4% (Rating: 264 MVA)
DFAX: 91.3%
Pre-project loading: 56.0%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Dona Vista - McQueeney 138.0 kV Line	Rebuild line with higher-capacity conductor	\$3,504,000	100% (\$3,504,000)
Villa Mar 1657 - McQueeney 138.0 kV	Rebuild line with higher-capacity conductor	\$5,107,000	83% (\$4,240,000)
Los Angeles 1617 - Hartselle 138.0 kV	Rebuild line with higher-capacity conductor	\$12,553,000	83% (\$10,428,000)
Villa Mar 1657 - Los Angeles 1617 Substation 138.0 kV L	Rebuild line with higher-capacity conductor	\$3,301,000	83% (\$2,740,000)
	Total allocated to project		\$20,912,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.